

is, if I am a data centre service provider, and I provide distributed data centre solution to multiple customers, I should be able to connect the different data centres securely for my customers. And I should be able to do this dynamically in case of changes required in the service or configuration. In each of these use cases multiple tenants in a data centre share the same physical resources (physical servers, physical storage and physical network). Each tenant is assigned its own logical resources (virtual machines, virtual storage and virtual networks). These logical resources are isolated from each other.

SDN primarily helps us in the following two ways:

1. **Cloud networking.** Helps generate private clouds for enterprises or service providers, builds and provides network infrastructure as a service (IaaS) and helps build virtual private clouds (VPCs) for cloud service providers (also known as MSP).

2. **Network function virtualisation (NFV) in service-provider networks.** This provides value-added services (VAS) for service-provider edge networks, such as business-edge networks, broadband-subscriber-management edge networks and mobile-edge networks.

SDN also helps in moving a number of network services to general-purpose hardware, eliminating the need for dedicated hardware for these applications. Examples of such applications are intrusion detection or prevention systems (IDS/IPS), deep packet inspection (DPI), caching and wide area network (WAN) optimisation in virtual machines instead of on physical hardware appliances.

Implementation

In its present state the structure of SDN is not very well defined. It looks like every vendor understands the problem they are trying to solve, but

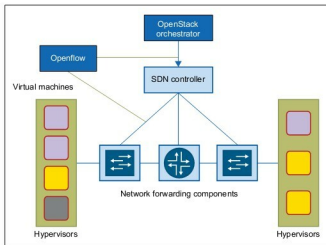


Fig. 3: Network forwarding components

every vendor has a different approach towards it. Some organisations are trying to build standards around the SDN.

One of the important aspects of SDN is to keep different components in the network physically separate and yet make them work together in cohesion. For example, if we take a high-performance integrated-service gateway, it is likely to have a firewall, IDP/IDS, routing and switching system, all built into one box. When we take all these services to an SDN, we have different software components doing the firewall, IDP routing and switching. Also, in SDN only the forwarding of the actual connectivity has to be provided by a dedicated physical hardware. All these different components are controlled by a master component called the SDN controller.

This brings in a challenge to making sure that all these different components work together and are controllable by a controller in the network. This requires a protocol to be developed for the communication among the different components in an SDN network.

To understand the very-high-level architecture of the SDN network please see Fig. 3.

Open Network Foundation (ONF) (<https://www.opennetworking.org>) is a non-profit industry consortium that is leading the advancement of SDN

and standardising critical elements of the SDN architecture, such as the OpenFlow protocol which structures communication between the control and data planes of supported network devices. OpenFlow is the first standard interface designed specifically for SDN, providing high-performance, granular traffic control across multiple vendors' network devices.

Network management

One of the most important aspects of SDN is seamless network management. The SDN aims at delivering an application as a whole, rather than each network service. The high-level network applications are formed with well-orchestrated configuration of different network services. This demands an implementation of a very powerful and open network management system (NMS). An NMS which translates a high-level business policy into a configuration of each of the different service layers, including routing, switching, security and virtual machine provisioning, is the core of an SDN network. Along with providing easy configurability it should also provide a clear visibility in the network and highlight problems, if any.

Standards

ONF is the main organisation driving the standardisation effort of the SDN. It mainly deals with defining the protocols and standardising the way services should be provided in a network. The main idea is to keep the network components and services interoperable, so that healthy competition is maintained, and customers can choose the best components from different vendors.

Strengths of SDN

There is a complete transition of the network architecture in SDN. In